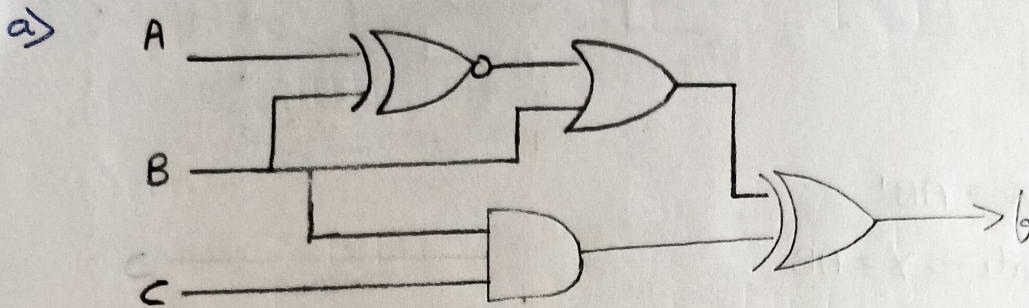


BOOLEAN ALGEBRA - ASSIGNMENT

I] Derive the Output expression 'b' for the below circuits and draw the Simplified Circuits.



Step 1 : XNOR gate

$$X = (A \oplus B)' = AB + A'B'$$

Step 2 : OR gate

$$Y = X + B$$
$$= (AB + A'B') + B$$

Using Absorption law,

$$AB + B = B$$

$$Y = B + A'B'$$

Using

$$X + YZ = (X + Y)(X + Z)$$

$$B + A'B' = (B + A')(B + B')$$

$$= (B + A')(1)$$

$$= \underline{\underline{A' + B}}$$

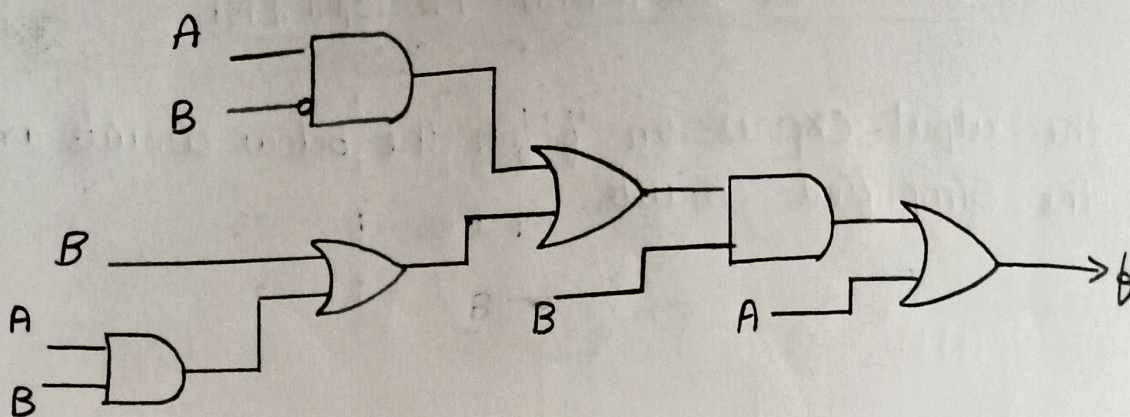
Step 3 := And gate

$$Z = BC$$

Step 4 := Final XOR gate

$$b = (A' + B) \oplus BC$$

b]



First gate: $x = AB'$

Bottom AND gate $\rightarrow Y = AB$

Bottom OR gate $\rightarrow B + AB = B$

Middle OR gate $\rightarrow AB' + B$

Using absorption, $AB' + B = A + B$

AND gate

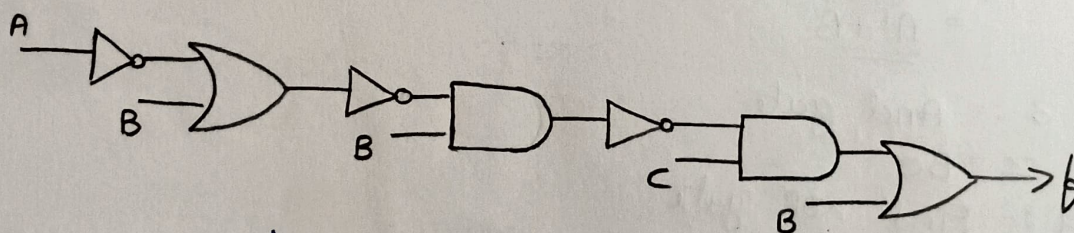
$$\begin{aligned} (A+B)B \\ = AB + B \\ = \underline{B} \end{aligned}$$

OR gate

$$B + A$$

$$\boxed{b = A + B}$$

c]



Inverter 1 $\Rightarrow A'$

OR gate $= A' + B$

Inverter 2 $= (A' + B)' = AB'$

And gate $= AB' \cdot B \Rightarrow A(B'B) = \underline{0}$

Inverter 3 $= 0' = \underline{1}$

And gate $= 1 \cdot C = C$

OR gate $= C + B$

$$\boxed{b = B + C}$$

2] Minimize the following Boolean Expressions Using Boolean postulates.

a) $\bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$$\bar{A}\bar{D}(CE + \bar{B}) + AB(CE + D)$$

$$F = \underline{\underline{\bar{A}\bar{D}(\bar{B} + CE) + AB(D + CE)}}$$

b) $F = \bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

Absorption law,

Since $X + XY = X$

$$\bar{A}C + \bar{A}C\bar{D} = \bar{A}C$$

Factor Common terms

$$= \bar{C}(\bar{A}\bar{B} + A\bar{B}) + \bar{A}C + A\bar{C}D + ABD$$

$$= \bar{B}\bar{C}(\bar{A} + A) + \bar{A}C + A\bar{C}D + ABD$$

$$= \bar{B}\bar{C} + \bar{A}C + A\bar{C}D + ABD$$

$$F = \underline{\underline{\bar{B}\bar{C} + \bar{A}C + A\bar{C}D + ABD}}$$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$

Absorption law

$$X + XY = X$$

$$F = \bar{A}\bar{C} + BC + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

Take $\bar{B}$ Common

$$F = \bar{A}\bar{C} + BC + \bar{B}(A + \bar{C}\bar{D}) + A\bar{C}\bar{D}$$

$$F = \underline{\underline{\bar{A}\bar{C} + BC + \bar{B}(A + \bar{C}\bar{D}) + A\bar{C}\bar{D}}}$$

$$d) F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 x_2 \bar{x}_3 x_4 + x_2$$

Use absorption law

$$x_2 + x_1 x_2 \bar{x}_3 x_4 = x_2$$

$$F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_2$$

Factor common term.

Take $\bar{x}_1$ common

$$F = \bar{x}_1 (\bar{x}_3 \bar{x}_4 + \bar{x}_2 x_4) + x_3 x_4 + x_2$$

Take $\bar{x}_1$ common

$$F = \bar{x}_1 (\bar{x}_3 \bar{x}_4 + \bar{x}_2 x_4) + x_3 x_4 + x_2$$

$$4) f(a, b, c, d) = \overline{\Pi m}(4, 6, 8, 10, 11, 12, 15) + \Sigma d(3, 5, 7, 9)$$

ab \ cd	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0 ₀	0 ₁	X ₃	0 ₂
$\bar{a}b$	1 ₄	X ₅	X ₇	1 ₆
$a\bar{b}$	1 ₁₂	0 ₁₃	1 ₁₅	0 ₁₄
ab	1 ₈	X ₉	1 ₁₁	1 ₁₀

$$f = \underline{a'b + ab' + b\bar{c}\bar{d} + acd}$$

$$b) f(a, b, c, d) = \overline{\Pi m}(0, 2, 4, 6, 7, 9) \cdot \Pi d(10, 11)$$

ab \ cd	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0 ₀	1 ₁	1 ₃	0 ₂
$\bar{a}b$	0 ₄	1 ₅	0 ₇	0 ₆
$a\bar{b}$	1 ₁₂	1 ₁₃	1 ₁₅	1 ₁₄
ab	1 ₈	0 ₉	X ₁₁	X ₁₀

$$f = \underline{(a+d) \cdot (a+b'+c') \cdot (a'+b+c')}$$